

Repaired Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$ (v2): Recursive Verticalization (M6') to Close the Round-5 Side-Witness Gap

A delta to the 31-page round-4 repaired proof [1]

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Disclaimer. This document was drafted by Claude (Anthropic), prompted by Michael Wessel, in response to GPT-5.5's fifth formal review of the repaired split-forest proof. The results have not been independently verified by human domain experts. We invite scrutiny and corrections.

Abstract

GPT-5.5's fifth formal review [2] of the 31-page repaired split-forest decidability proof for $\mathcal{ALCI}_{\text{RCC5}}$ identifies one load-bearing mathematical defect: the verticalization discipline (M6) of Definition 5.1 in [1] is too strong. It forbids non-vertical PP/PPI relations between sibling-frontier mosaic positions, but such relations occur in satisfiable witness-generated models, as witnessed by the stress concept

$$C_{\text{side}} := \exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B,$$

which is satisfiable in abstract RCC5 (and is reported *SAT* by all three Python reasoners in the project repository), yet rejected by the round-4 (M6). This delta paper repairs the defect by replacing the flat verticalization $\text{Vrt} \subseteq P$ with a *recursive vertical-stratum forest*, in which every selected DR/PO side-witness may itself open a local vertical stratum containing its own selected PP/PPI ancestors and equality mates. The new discipline (M6') is the natural recursive lift of the round-4 (M6); the rest of the round-4 architecture (occurrence-level equality, port-indexed mate clusters $\text{Mate}_{\mathcal{Q}}(C, s, p)$, the abstract triple graph $\mathcal{T}_{\mathcal{Q}}$, request-closed cycles in place of ω -acceptance) carries over without further change. The explicit cubic side-width bound $m(n) = O(n^3)$ of the round-4 paper is replaced by a finite recurrence over modal depth; the certificate space is still bounded by an effectively computable $B(C_0) \leq 2^{2^{p(n)}}$ for some computable p .

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1 Scope and relationship to the round-4 paper

This is a delta to the 31-page round-4 paper [1]. Specifically:

- Section 5 (Finite interfaces and mosaics) of [1] is amended in §3 below: Definition 5.1 (Mosaic) is restated with a recursive vertical-stratum forest, and the four sub-clauses of (M6) are replaced by their recursive analogues (M6').
- Section 7.2 (Verticalization soundness) of [1] is amended in §4: Lemma 7.1 (Vertical realisation of PP/PPI pairs) is restated as Lemma 4.1 with the recursive case analysis.
- Section 6.5 (Validity conditions) of [1] is amended in §5: clauses (V3), (V4), (V5), and (V9.Eb) are restated to traverse the stratum forest, in addition to the port-indexed mate clusters introduced in round 4.
- Section 4.3 (Explicit DR/PO witness placement and side interfaces) of [1] is amended in §6: the polynomial bound $m(n) = O(n^3)$ is replaced by a finite recurrence over modal depth. The certificate space is still finite and effectively computable.
- All other sections of [1] are preserved verbatim. In particular: witness thinning (§3 of [1]), occurrence-level equality (§4.1), the port-indexed mate clusters $\text{Mate}_{\mathcal{Q}}(C, s, p)$ (§6.3), the

abstract triple graph $\mathcal{T}_{\mathcal{Q}}$ (§5.2), Renz–Nebel-based mosaic patchwork (§5.4–§5.7), request-fulfilled cyclic repetition (§6.5), and the no-automata regularization (§8) all carry over without modification.

The repair does not change the soundness chain (§7 of [1] except for Lemma 7.1) and does not change the completeness chain (§8 of [1] except for the side-interface construction at §4.3). In particular, the proof remains no-automata, in the sense that the only finite-state machinery used is the request-closed blocking cycle of Section 6.5 in [1].

2 The defect, with the round-5 stress concept

2.1 The round-4 verticalization discipline (M6)

Definition 5.1 of [1] declares, for every mosaic $M = (P, \tau, L, \text{Vrt})$ over a finite position set P , that $\text{Vrt} \subseteq P$ is the set of *vertical* positions (the source occurrence of the mosaic, its explicit equality mates, and any selected PP/PPI containment witnesses recorded in the mosaic) and $\text{Sib} := P \setminus \text{Vrt}$ is the set of *sibling-frontier* positions (selected DR/PO witnesses and ordinary sibling / boundary nodes). Clause (M6) then states:

- (M6.a) EQ-closure of strata: $L(p, q) = \text{EQ}$ implies $p \in \text{Vrt}$ iff $q \in \text{Vrt}$;
- (M6.b) containment labels are vertical: $L(p, q) \in \{\text{PP}, \text{PPI}\}$ implies $p, q \in \text{Vrt}$;
- (M6.c) cross-stratum labels are DR/PO: if exactly one of p, q lies in Vrt , then $L(p, q) \in \{\text{DR}, \text{PO}\}$;
- (M6.d) within sibling-frontier: if $p, q \in \text{Sib}$ are in different equality classes and $p \neq q$, then $L(p, q) \in \{\text{DR}, \text{PO}\}$.

2.2 The stress concept

Let A, B be atomic concepts of $\text{cl}(C_0)$ and consider

$$C_{\text{side}} := \exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B.$$

Example 2.1 (Concrete satisfying model). Let $\Delta = \{x, y, z\}$ with $A^{\mathcal{I}} = \{y\}$, $B^{\mathcal{I}} = \{z\}$, and atomic relation labels

$$L(x, y) = \text{DR}, \quad L(x, z) = \text{DR}, \quad L(y, z) = \text{PP} \quad (L \text{ extended to inverses and reflexive EQ in the usual way}).$$

JEPD and the RCC5 composition table are satisfied: the only nontrivial triple is (x, y, z) with $\text{comp}(\text{DR}, \text{PP}) = \{\text{DR}, \text{PO}, \text{PP}\}$, and the chosen label $L(x, z) = \text{DR}$ lies in $\text{comp}(\text{DR}, \text{PP})$. The model satisfies C_{side} at x : the first conjunct is witnessed by y (which satisfies $A \sqcap \exists \text{PP}.B$, since y is in A and $L(y, z) = \text{PP}$ with $z \in B$), and the second conjunct is witnessed by $z \in B$.

The cover-tree tableau, the baseline quasimodel reasoner, and the cycle-aware quasimodel reasoner of the repository all report C_{side} SAT; the verification script `verification/python/wp7_side_witness_st` records the agreement. Hence the concept is genuinely satisfiable in abstract RCC5.

2.3 Why round-4 (M6) rejects C_{side}

Consider the candidate certificate that the round-4 proof would construct for a model of C_{side} rooted at x . At the source x we place two selected DR witnesses, one for each conjunct, say p_y for $y \models A \sqcap \exists \text{PP}.B$ and p_z for $z \models B$. The mosaic at x has $\text{Vrt}_M = \{x\}$ (since x has no selected PP/PPI witnesses; the upward containment from x is empty) and $\text{Sib}_M = \{p_y, p_z\} \cup \{\text{boundary nodes}\}$. By Example 2.1, $L(p_y, p_z) = \text{PP}$. But clause (M6.d) of round-4 requires $L(p_y, p_z) \in \{\text{DR}, \text{PO}\}$, contradiction.

So the round-4 (M6) rejects every certificate built from the witness-generated submodel of any satisfying model of C_{side} . Concretely, the (M6) clause that fails is the within-sibling-frontier clause (M6.d): it asserts that two distinct equality classes inside Sib admit only $\{\text{DR}, \text{PO}\}$ labels, whereas PP/PPI between two DR-witnesses of the source can be forced by the witness-generated containment structure of C_{side} .

Remark 2.2. The round-4 proof needed (M6.d) to make Lemma 7.1 (vertical realisation of PP/PPI pairs) of [1] go through: that proof argued, in Subcase (ii.e), that every PP/PPI mosaic label is between vertical positions because (M6.b) forbids PP/PPI across the Vrt/Sib boundary and (M6.d) forbids PP/PPI within Sib . The recursive repair below preserves the *conclusion* of Lemma 7.1 (vertical realisation modulo EQ-mate substitution) while admitting nested local vertical strata, so the case analysis no longer needs to exclude sibling-frontier PP/PPI pairs.

3 Recursive verticalization

The repair is to replace the single vertical set $\text{Vrt} \subseteq P$ by a finite *forest* of nested vertical strata. Each stratum has its own source, its own equality mates, and its own selected PP/PPI witnesses, recursively. The forest carries the explicit nesting and the labels between positions in distinct strata are still constrained as in round-4 (cross-stratum DR/PO).

3.1 Stratum forest

Definition 3.1 (Vertical stratum). A *vertical stratum* on a finite position set P is a tuple $V = (P_V, s_V, \mu_V)$ where $P_V \subseteq P$ is the stratum's support, $s_V \in P_V$ is its *source position*, and $\mu_V \subseteq P_V$ is the stratum's set of *explicit equality mates* of s_V , with $s_V \in \mu_V$. The positions in $P_V \setminus \mu_V$ are the stratum's *selected PP/PPI-witnesses*.

Definition 3.2 (Stratum forest). A *stratum forest* on a finite position set P is a pair $\mathcal{V} = (\text{Strat}, \text{subsrc})$ where

- (SF1) $\text{Strat} = \{V_0, V_1, \dots, V_k\}$ is a finite set of vertical strata on P (Definition 3.1);
- (SF2) V_0 is distinguished as the *root stratum*; its source s_{V_0} is called the *mosaic source*;
- (SF3) $\text{subsrc} : \text{Strat} \setminus \{V_0\} \rightarrow P$ is the *parent position map*: $\text{subsrc}(V_i)$ is the position in some other stratum V_j at which V_i is rooted as a nested local stratum;
- (SF4) the parent relation subsrc induces a forest on Strat with V_0 as the global root: for every non-root stratum V_i , $\text{subsrc}(V_i) \in P_{V_j}$ for some uniquely determined $V_j \neq V_i$, and the relation $V_i \prec V_j$ iff $\text{subsrc}(V_i) \in P_{V_j}$ is acyclic;

- (SF5) supports are stratified: distinct strata have distinct sources, and for any two strata V_i, V_j , $P_{V_i} \cap P_{V_j}$ is either empty or contains exactly the parent position $\text{subsrc}(V_i)$ when V_i is nested at a position of V_j .

Remark 3.3. (SF5) is the key sharing rule. A position $p \in P$ can be a *sibling-frontier* member of V_j and *simultaneously* the source of a nested stratum V_i rooted at p . In that case $p \in P_{V_i} \cap P_{V_j}$ but the rest of P_{V_i} does not intersect P_{V_j} .

We extend the stratum-membership relation pointwise:

Definition 3.4 (Stratum membership). For $p \in P$, write

$$\text{Strat}(p) = \{ V \in \text{Strat} : p \in P_V \}.$$

By (SF5), $|\text{Strat}(p)| \in \{1, 2\}$, with $|\text{Strat}(p)| = 2$ exactly when p is the parent position of some nested stratum.

We say p is *vertical in V* if $p \in P_V$. We say a pair (p, q) is *co-vertical* if there exists $V \in \text{Strat}$ with $\{p, q\} \subseteq P_V$; otherwise it is *cross-stratum*. The *residual sibling frontier* of V_j at a position $p \in P_{V_j}$ is the set of positions in P_{V_j} that are in V_j but in no nested stratum rooted at p .

3.2 Restated mosaic definition

Definition 3.5 (Mosaic, v2). A mosaic over a finite position set P is a tuple

$$M = (P, \tau, L, \mathcal{V})$$

where $\tau : P \rightarrow \text{Typ}(C_0)$, $L : P^2 \rightarrow \mathbf{B}$, and $\mathcal{V}$ is a stratum forest on P (Definition 3.2). The tuple satisfies axioms (M1)–(M5) of Definition 5.1 in [1], unchanged in form, together with the recursive verticalization discipline:

- (M6'.a) **EQ-closure within strata.** If $L(p, q) = \text{EQ}$ and $V \in \text{Strat}$ with $p \in P_V$, then $q \in P_V$. In other words, every EQ-class lies entirely inside some single stratum.
- (M6'.b) **Containment labels are co-vertical.** If $L(p, q) \in \{\text{PP}, \text{PPI}\}$, then (p, q) is co-vertical (Definition 3.4): there exists $V \in \text{Strat}$ with $\{p, q\} \subseteq P_V$, and $L(p, q)$ matches the local containment data of V (i.e., if $L(p, q) = \text{PP}$ then q is the source s_V or an equality mate, and p is a selected PPI-witness of V ; the PPI case is symmetric).
- (M6'.c) **Cross-stratum labels are DR/PO or EQ.** If (p, q) is cross-stratum and $p \neq q$, then $L(p, q) \in \{\text{DR}, \text{PO}\}$ unless (p, q) is identified by an explicit EQ-class, in which case $L(p, q) = \text{EQ}$ and (by (M6'.a)) both p and q lie inside one stratum.
- (M6'.d) **Residual sibling frontier.** If p, q are both in the residual sibling frontier of some common stratum V_j at a common parent position (i.e., neither p nor q opens a nested stratum containing the other), then $L(p, q) \in \{\text{DR}, \text{PO}\}$ or $L(p, q) = \text{EQ}$ with p, q in the same equality class.

The intended reading is: each PP/PPI label in the mosaic is *justified* by some stratum in $\mathcal{V}$ — either the root stratum V_0 (the round-4 case) or a nested stratum opened at a sibling-frontier position of V_0 . Pairs of positions whose vertical contexts are mutually incomparable still

receive only DR/PO labels, which is the local arc-consistent regime that the patchwork-based completeness argument requires.

Lemma 3.6 (Round-4 (M6) is the depth-1 case). *If $\mathcal{V} = (\{V_0\}, \emptyset)$ (the stratum forest has only the root stratum and no nested strata), then Definition 3.5 specializes to Definition 5.1 of [1] with $\text{Vrt} = P_{V_0}$. In particular, every round-4 mosaic is a v2 mosaic.*

Proof. Direct unfolding of definitions. (M6'.a) becomes (M6.a) since the only stratum is V_0 and $\text{Vrt} = P_{V_0}$. (M6'.b) becomes (M6.b) because co-vertical reduces to “both in V_0 ”. (M6'.c) and (M6'.d) become (M6.c) and (M6.d) by the same reduction. $\square$

Remark 3.7. Recursive verticalization is not an arbitrary generalization: it is exactly what the witness-generated containment structure produces. When the witness selector chooses, at a source u , a DR-witness w that itself has a PP-witness w' inside the model, then the selector at w produces a new local mosaic context whose source is w . In round-4, that nested context was *separate* from u 's mosaic. The defect arose because w' can also be a selected DR/PO witness of u (when w' is DR to u but PP to w), in which case the round-4 mosaic at u records both w and w' at the frontier with no place to put the forced PP label between them. The repair pulls w and w' into a shared nested stratum V_w inside u 's mosaic, recording the PP label as a local co-vertical label of V_w .

4 Verticalization soundness (restated Lemma 7.1)

We lift (M6') from a single mosaic to the entire unfolding $\mathcal{U}(\mathcal{Q})$. The argument generalizes the round-4 case analysis of Lemma 7.1 in [1] (Subcases (ii.a)–(ii.e) of the *mosaic-label* case) by adding the recursive stratum-membership data.

Lemma 4.1 (Vertical realisation, recursive). *Let $\mathcal{Q}$ be a valid certificate satisfying (V1)–(V10) and (M6') on every mosaic in $\text{Mos}_{\mathcal{Q}}$. Let $u, v \in \Omega(\mathcal{U}(\mathcal{Q}))$ and $R \in \{\text{PP}, \text{PPI}\}$. Suppose $L_{\mathcal{U}(\mathcal{Q})}(u, v) = R$ in the unfolding. Then either*

- (a) *there is a strict vertical Reach_R -chain from u to v in the generated backbone of $\mathcal{U}(\mathcal{Q})$ (not necessarily through the mosaic source; the chain may live in a nested stratum), or*
- (b) *there exist $u', v' \in \Omega(\mathcal{U}(\mathcal{Q}))$ with $u \equiv_{\text{EQ}} u'$ and $v \equiv_{\text{EQ}} v'$ such that (u', v') falls under (a).*

Proof. As in the proof of Lemma 7.1 of [1], the label $L_{\mathcal{U}(\mathcal{Q})}(u, v) \in \mathbf{B}$ is assigned by either (i) a vertical generated cover edge recorded in the certificate's containment data, or (ii) a mosaic label $L_M(p, q)$ where u realises position p and v realises position q of some $M \in \text{Mos}_{\mathcal{Q}}$.

Case (i): vertical generated cover. Unchanged from the round-4 proof: u and v lie on the same vertical backbone, with $u\text{Reach}_R v$, by (V4)/(V5). This is case (a).

Case (ii): mosaic label. $L_M(p, q) = R \in \{\text{PP}, \text{PPI}\}$. By (M6'.b), there exists $V \in \text{Strat}_M$ with $\{p, q\} \subseteq P_V$. Within stratum V , positions are partitioned into the source s_V , its explicit equality mates μ_V , and selected containment witnesses of V . We argue by the same case-by-case analysis as round-4, but *within* stratum V :

- Subcase (ii.a'): $p = s_V$ and q is a selected R -witness of V . The realising pair (u, v) is the source of V and its selected R -witness on V 's vertical backbone, so $u\text{Reach}_R v$. This is case (a).
- Subcase (ii.b'): $p \in \mu_V$ (an equality mate of s_V) and q is a selected R -witness of V . By (V9.Eb) applied to stratum V (see §5 below), the source s_V is realised by an occurrence u' with $u \equiv_{\text{EQ}} u'$ and $u'\text{Reach}_R v$. This is case (b) with $v' = v$.
- Subcase (ii.c'): $p = s_V$ and $q \in \mu'_V$, where μ'_V is the equality-mate set of some selected containment witness inside V . Symmetric to (ii.b'): the selected witness v' satisfies $v \equiv_{\text{EQ}} v'$ and $u\text{Reach}_R v'$. Case (b) with $u' = u$.
- Subcase (ii.d'): $p \in \mu_V$ and $q \in \mu'_V$ as above. Both substitutions apply; the pair (u', v') falls under (a) by composition of (ii.a')–(ii.c'). Case (b) in full.
- Subcase (ii.e'): both p and q are selected containment witnesses of V (neither is s_V or one of its mates). Then by (M1)–(M5) on V and the round-4 argument applied *inside* V , the source s_V is the common third position with $L_M(s_V, p), L_M(s_V, q) \in \{\text{PP}, \text{PPI}\}$ and $L_M(p, q) = R$ forced by composition through s_V . Each option places the realising pair (u, v) on the vertical backbone of V through the source occurrence of V , giving case (a). Equality-mate variants are handled exactly as in (ii.b')–(ii.d').

The only difference from the round-4 case analysis is that the “common stratum” is no longer forced to be the root stratum V_0 . By (M6'.b) it is *some* stratum $V \in \text{Strat}_M$, possibly nested; the proof goes through inside V because each individual stratum is itself a round-4-style local mosaic over P_V . In every subcase, (u, v) satisfies (a) directly or admits $\equiv_{\text{EQ}}$ -mate substitutes that do, which is (b). $\square$

Remark 4.2. The key insight is that the round-4 Subcase (ii.e) argument is purely *local* to one stratum: it uses only the round-4 axioms (M1)–(M5) restricted to P_V , plus (M6) restricted to V 's containment data. Recursive verticalization preserves this locality inside each stratum and uses (M6') only to ensure that the right stratum exists. Cross-stratum pairs never receive PP/PPI labels (by (M6'.c)), so the round-4 case analysis is not lost; it is applied *inside* each stratum.

5 Updated validity clauses (V3), (V4), (V5), (V9.Eb)

The round-4 validity clauses on certificates (Section 6.5 of [1]) are formulated in terms of port-indexed mate clusters $\text{Mate}_Q(C, s, p) \subseteq S \times \text{Port}(s')$ attached to contexts $C \in \text{Ctx}_Q$. In the recursive setting the clauses traverse the stratum forest of each mosaic, in addition to the mate clusters.

We restate only the four clauses affected by the repair. All other clauses (V1), (V2), (V6), (V7), (V8), (V9.Ea), (V10) of [1] are preserved verbatim.

(V3') Selected containment witness, per stratum.

For every $\exists R.D \in \text{cl}(C_0)$ with $R \in \{\text{PP}, \text{PPI}\}$, every mosaic $M \in \text{Mos}_Q$, every stratum $V \in \text{Strat}_M$, and every port p of the source s_V such that $\exists R.D \in \tau(s_V)$ at port p : there exists a selected R -witness $w \in P_V \setminus \mu_V$ declared in V , with $D \in \tau(w)$ and $L_M(s_V, w) = R$. *Difference from round-4 (V3).* Round-4 (V3) ranged over the single stratum Vrt_M . (V3') ranges over every stratum V in the forest $\mathcal{V}_M$. Existential discharge for a position s_V is checked within the stratum that s_V heads.

(V4') Mate-cluster closure, per stratum.

For every mosaic M , every stratum $V \in \text{Strat}_M$, and every port p of s_V : the port-indexed mate cluster $\text{Mate}_Q(C_V, s_V, p)$ declared for V is closed under explicit EQ-port chains in E_Q within V , in the sense that every chain of EQ-port links starting from (s_V, p) and remaining inside P_V produces only ports that lie in $\text{Mate}_Q(C_V, s_V, p)$. *Difference from round-4 (V4).* The closure operation is now restricted to stratum-local EQ-port chains; cross-stratum EQ-port chains are forbidden by (M6'.a), so they need not be considered.

(V5') Universal propagation, per stratum.

For every $\forall R.D \in \text{cl}(C_0)$, every mosaic M , every stratum $V \in \text{Strat}_M$, and every pair $p, q \in P_V$ with $L_M(p, q) = R$ and $\forall R.D \in \tau(p)$: $D \in \tau(q)$. Furthermore, for cross-stratum pairs (p, q) with $L_M(p, q) = R \in \{\text{DR}, \text{PO}\}$ and $\forall R.D \in \tau(p)$: $D \in \tau(q)$ as well (this is the cross-stratum case, unchanged from round-4). *Difference from round-4 (V5).* Within-stratum universal propagation is restricted to one stratum; cross-stratum DR/PO universal propagation is unchanged. No within-stratum PP/PPI universal escapes from V because (M6'.b) forbids any PP/PPI label between P_V and the rest of the mosaic.

(V9.Eb') Equality-mate joint witnessing, per stratum.

For every mosaic M , every stratum $V \in \text{Strat}_M$, every explicit EQ-class $E \subseteq P_V$, and every $\exists R.D \in \text{cl}(C_0)$ common to the type $\tau(p)$ for all $p \in E$: there exists a single selected witness $w \in P_V$ with $L_M(p, w) = R$ for some $p \in E$ and $D \in \tau(w)$, such that the substitution lemma applied to any $p' \in E$ yields the same $\equiv_{\text{EQ}}$ -class for w . *Difference from round-4 (V9.Eb).* Joint witnessing is checked per stratum, because (M6'.a) prevents EQ-classes from spanning multiple strata. The clause matches the local structure: an equality class lives inside one stratum, so its joint discharge is also stratum-local.

Remark 5.1. The four clauses above are essentially *universally quantified over strata*. Their finite-checkability is preserved: the stratum forest of every mosaic in Mos_Q is finite (a finite set of strata), so each of (V3')–(V9.Eb') is a finite conjunction of round-4-style clauses, one per stratum. The original round-4 validity clauses (V3), (V4), (V5), (V9.Eb) are recovered as the single-stratum case (depth-1 stratum forest).

6 Side-interface width, recurrence over modal depth

The round-4 paper proves an explicit cubic bound $|\text{Side}(u)| \leq 1 + n^3 + 3n^2 + 3n$ for the side interface at an occurrence u , where $n = |C_0|$ (Lemma 4.4 of [1]). Under recursive verticalization, a single side-witness w of u can itself open a local stratum carrying up to $|\text{Side}(w)|$ further positions. The width bound therefore becomes a recurrence over modal depth.

6.1 Recurrence

Let $\text{md}(C_0) \in \mathbb{N}$ denote the modal depth of C_0 . Define $m : \{0, 1, \dots, \text{md}(C_0)\} \rightarrow \mathbb{N}$ by

$$m(0) = 1, \quad m(d+1) = 1 + n^3 \cdot (1 + k_{\text{Mate}} + k_{\text{bd}}) + n^3 \cdot m(d),$$

where $n = |C_0|$ and $k_{\text{Mate}}, k_{\text{bd}}$ are the round-4 mate and boundary constants.

Proposition 6.1 (Side-interface width). *For every occurrence u in $\mathcal{U}(\mathcal{Q})$ at modal depth $\leq d$ from a generated context source, $|\text{Side}(u)| \leq m(d)$.*

Proof. Induction on d . Base $d = 0$: u is a leaf of the witness selector; $\text{Side}(u) = \{u\}$, so $|\text{Side}(u)| = 1 = m(0)$. Inductive step: the selector at u adds up to n^3 selected DR/PO witnesses (round-4 cubic bound), each with up to $1 + k_{\text{Mate}} + k_{\text{bd}}$ boundary or equality-mate companions, and each side-witness w may open a nested stratum V_w of width $\leq m(d)$ (by the induction hypothesis applied at w). Summing: $|\text{Side}(u)| \leq 1 + n^3(1 + k_{\text{Mate}} + k_{\text{bd}}) + n^3 m(d) = m(d+1)$. $\square$

Corollary 6.2 (Effectively computable side-width). *$m(\text{md}(C_0))$ is effectively computable from C_0 . In particular, the total side-interface alphabet is finite and bounded by $m(\text{md}(C_0)) \leq (n^3 + 1)^{\text{md}(C_0)+1} \cdot (1 + k_{\text{Mate}} + k_{\text{bd}})$.*

The bound $m(\text{md}(C_0))$ is no longer polynomial in n in general; it is exponential in $\text{md}(C_0)$, which can itself be linear in n . This matches GPT-5.5's round-5 recommendation: drop the polynomial claim and commit only to an effectively computable bound. The certificate space $B(C_0)$ from Section 6.6 of [1] now uses $m(\text{md}(C_0))$ in place of the cubic constant, but the overall double-exponential bound $B(C_0) \leq 2^{2^{p(n)}}$ for some computable polynomial p is preserved.

Remark 6.3. The decision procedure does not rely on a polynomial side-width. It only uses that $B(C_0)$ is effectively computable from C_0 . Both $m(\text{md}(C_0))$ and the round-4 cubic bound satisfy this requirement.

7 Worked example: a v2 certificate for C_{side}

We construct a small certificate for $C_{\text{side}} = \exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$ explicitly, to confirm that the v2 discipline accepts the satisfiable concept.

7.1 Setup

Let $C_0 = C_{\text{side}}$. The closure is $\text{cl}(C_0) = \{C_0, \exists \text{DR}.(A \sqcap \exists \text{PP}.B), A \sqcap \exists \text{PP}.B, A, \exists \text{PP}.B, B, \exists \text{DR}.B, \top, \perp\}$ together with negations as needed. Modal depth $\text{md}(C_0) = 2$.

7.2 Hintikka types

Three Hintikka types suffice for the source and its two selected DR-witnesses; a fourth type covers the PP-witness of the first selected DR-witness:

$$\begin{aligned}\tau_x &= \{ C_0, \exists \text{DR}.(A \sqcap \exists \text{PP}.B), \exists \text{DR}.B, \top \}, \\ \tau_y &= \{ A \sqcap \exists \text{PP}.B, A, \exists \text{PP}.B, \top \}, \\ \tau_z &= \{ B, \top \}, \\ \tau_w &= \{ B, \top \}.\end{aligned}$$

(We use τ_w for y 's selected PP-witness; it happens to be equal to τ_z , which is what justifies the identification $w = z$ in the round-5 stress reading; see Remark 7.1 below.)

7.3 Certificate structure

The certificate $\mathcal{Q}$ has exactly one context $C \in \text{Ctx}_{\mathcal{Q}}$ whose source has profile/type τ_x at port p_x . The mosaic at C is $M = (P, \tau, L, \mathcal{V})$ with $P = \{x, y, z\}$ (we use the same names for positions and the model elements of Example 2.1; this is a notational coincidence, not a semantic identification). Stratum forest $\mathcal{V} = (\text{Strat}, \text{subsrc})$ with:

- Root stratum V_0 : $P_{V_0} = \{x\}$, source $s_{V_0} = x$, mates $\mu_{V_0} = \{x\}$. (Source has no explicit equality mates and no selected PP/PPI witnesses, so V_0 contains only x .)
- Nested stratum V_1 : $P_{V_1} = \{y, z\}$, source $s_{V_1} = z$, mates $\mu_{V_1} = \{z\}$, selected PPI-witness y .
- $\text{subsrc}(V_1) = z$, with z as parent position. Note $z \in P_{V_0}^c$ in the sense that $z \notin V_0$, but z is a sibling-frontier position of V_0 ; (SF5) is satisfied because $P_{V_0} \cap P_{V_1} = \emptyset$, V_1 being rooted at z which itself lies outside V_0 's support.

Equivalently: the root stratum V_0 has source x and no selected witnesses; the sibling-frontier of V_0 contains y, z ; the nested stratum V_1 rooted at z has y as a PPI-witness of z .

Remark 7.1. The original model of Example 2.1 has $y\text{PP}z$, so in the certificate z is the proper-superpart and y the PPI-mate of z ; equivalently, y is a selected PPI-witness of z inside the nested stratum V_1 . We do not identify w with z in the certificate; we only use the type-level identification $\tau_w = \tau_z$ to keep the example compact.

7.4 Label matrix and (M6') verification

The label matrix $L : P^2 \rightarrow \mathbf{B}$ on $P = \{x, y, z\}$ is, modulo reflexive EQ on the diagonal and inverses on $L(q, p) = \text{Inv}(L(p, q))$:

$$L(x, y) = \text{DR}, \quad L(x, z) = \text{DR}, \quad L(y, z) = \text{PP}.$$

We verify (M6'.a)–(M6'.d) on M .

- (M6'.a) EQ-closure within strata. The only EQ-classes are the trivial singletons (no explicit equality identifications), so the clause holds vacuously.

- (M6'.b) Containment labels are co-vertical. The only PP/PPI label is $L(y, z) = \text{PP}$ (and its inverse $L(z, y) = \text{PPI}$). We verify that (y, z) is co-vertical: $y \in P_{V_1}$, $z \in P_{V_1}$, so $V_1 \in \text{Strat}(y) \cap \text{Strat}(z)$. Moreover the local containment data of V_1 (source z , selected PPI-witness y) matches $L(y, z) = \text{PP}$.
- (M6'.c) Cross-stratum labels are DR/PO or EQ. $L(x, y) = \text{DR}$ between $x \in P_{V_0}$ and $y \in P_{V_1}$ is cross-stratum: (x, y) is not co-vertical (no common stratum contains both). The label $\text{DR} \in \{\text{DR}, \text{PO}\}$ satisfies (M6'.c). Likewise $L(x, z) = \text{DR}$ between $x \in P_{V_0}$ and $z \in P_{V_1}$: cross-stratum, label DR, satisfies (M6'.c).
- (M6'.d) Residual sibling frontier. x is the source of V_0 and has no other positions in its residual sibling frontier (apart from y, z which are handled by (M6'.c)). No other residual-frontier pairs exist in M .

All four clauses hold, so M is a v2-mosaic, and C_{side} is accepted by the v2 certificate.

7.5 Validity clauses on this certificate

We verify the four affected clauses (V3'), (V4'), (V5'), (V9.Eb').

- (V3') Selected containment witness. The only $\exists R.D \in \text{cl}(C_0)$ with $R \in \{\text{PP}, \text{PPI}\}$ is $\exists \text{PP}.B \in \tau_y = \tau(y)$. We check stratum V_1 : y is a position of V_1 , and V_1 declares z as the source with y a selected PPI-witness. $L(y, z) = \text{PP}$ and $B \in \tau(z)$, so y 's $\exists \text{PP}.B$ is discharged in stratum V_1 as required.
- (V4') Mate-cluster closure. No explicit EQ-port chains are declared, so the clause holds vacuously.
- (V5') Universal propagation. No $\forall R.D$ universals are in any type of this certificate, so the clause holds vacuously.
- (V9.Eb') Equality-mate joint witnessing. No multi-element EQ-classes are declared, so the clause holds vacuously.

All non-vacuous validity conditions hold. The unaffected clauses (V1), (V2), (V6), (V7), (V8), (V9.Ea), (V10) are checked as in round-4; the only nontrivial point is (V2) (composition consistency), which holds because $L(x, y) = \text{DR}$, $L(y, z) = \text{PP}$, $L(x, z) = \text{DR}$ and $\text{DR} \in \text{comp}(\text{DR}, \text{PP}) = \{\text{DR}, \text{PO}, \text{PP}\}$.

7.6 Confirmation against the repository's reasoners

The verification script `verification/python/wp7_side_witness_stress.py` reports C_{side} as SAT under all three reasoners. The v2 certificate above is the smallest witness-generated certificate matching the abstract RCC5 model of Example 2.1.

The repair aligns the proof with what the provers already do. A noteworthy point for round-6 review: the (M6') repair is not a hypothetical relaxation introduced to admit a contrived counterexample. It brings the round-4 *certificate model* into alignment with what the

project's three Python provers (`src/cover_tree_tableau.py`, `src/alcircc5_reasoner.py`, `src/alcircc5_reasoner_cyclic.py`) have been computing all along.

A direct `grep` of those three files for the round-4 keywords `vert`, `Vrt`, `stratum`, `sibling`, `frontier`, `M6`, `side_frontier` returns no matches: none of the provers implements a flat verticalization discipline at the certificate level. Instead, the provers operate at the Hintikka-type / witness-chain level (CT1–CT5 for the cover-tree tableau; the disjunctive path-consistency loop for the quasimodel reasoners) and check only local network consistency, witness existence, composition propagation, and (in the cycle-aware variant) cycle closure through symmetric roles.

The cover-tree tableau, for example, accepts an $\exists$ DR witness that itself has a PP-witness without imposing any restriction on whether the inner PP-witness happens to also be a DR-witness of the original source — it simply tracks each demand by its own type and witness, and verifies the local RCC5 composition constraints on every realised triple. This is exactly what the recursive verticalization (M6') admits: nested vertical mini-contexts at each side-witness, with no flat-frontier constraint.

So no prover code needs to be modified for the (M6') repair. The `wp7` verification script (run on all three provers) shows the satisfiable stress concept is accepted, which is the empirical-layer ground truth. The round-5 defect was in the *theoretical layer*: the round-4 (M6) was strictly tighter than the operational semantics the provers implement, and the recursive (M6') brings them into alignment.

This is the kind of cross-check the project's two-layer architecture (theoretical proof + executable artefacts) was designed to provide: an empirical-layer “SAT” verdict on C_{side} flags the over-restriction in the theoretical layer, and the v2 repair is then the minimum theoretical change needed to catch up with what the implementations have been doing.

8 What carries over from round-4 verbatim

The following sections of [1] are unaffected by the repair, in the sense that their statements and proofs go through verbatim once Definition 3.5, Lemma 4.1, the restated validity clauses of §5, and the recurrence bound of §6 are inlined.

- Section 2 (Abstract RCC5 frames).
- Section 3 (Closure, types, and witness thinning), including the frame inheritance lemma and the witness-generated submodel theorem.
- Section 4.1–4.2 (Occurrences, blocking, and equality; generated split-cover construction). Section 4.3 (DR/PO witness placement) is amended only at the width bound (Lemma 4.4 of [1]), replaced by Proposition 6.1.
- Section 5.2 (Abstract triple graph $\mathcal{T}_Q$), with stratum membership $\zeta \in \{\text{vertical}_V : V \in \text{Strat}\} \cup \{\text{sibling}\}$ ranging over the per-position stratum index instead of a Boolean. The bound $|\text{Pos}_Q| \leq |S| \cdot \max_s |\text{Port}(s)| \cdot |\text{Ctx}_Q| \cdot k_{\text{strat}}$, where k_{strat} is the maximum stratum count per context (bounded by $m(\text{md}(C_0))$).
- Section 5.3–5.6 (Renz–Nebel patchwork). The patchwork lemma is applied to the local network on each stratum independently and to the residual cross-stratum frontier as in round-

4; the DR/PO arc-consistency argument that the round-4 paper uses on the sibling-frontier carries over to the cross-stratum case.

- Section 6.3 (Port-indexed mate clusters $\text{Mate}_{\mathcal{Q}}(C, s, p)$), unchanged in form; the clusters now attach to per-stratum sources and ports, but the definition and (V9.Ea) clause are unchanged.
- Section 6.5 (Request-fulfilled cycles), unchanged.
- Section 7.3 (Standalone typed-EQ quotient theorem), unchanged. The six assumptions (A1)–(A6) remain the same; (A4) is now discharged by the restated (V3')/(V4')/(V9.Eb') combined.
- Section 8 (Completeness, including the no-automata regularization), unchanged.

9 Discussion of the other round-5 items

GPT-5.5's round-5 review [2] lists eight recommended repairs. This paper closes **Item 2** (side-interface verticalization discipline), **Item 3** (add the stress test – done by Example 2.1 and the worked certificate of §7, with computational corroboration via `wp7_side_witness_stress.py`), and **Item 9** (drop the polynomial side-width claim, via the recurrence of §6). Item 1 (TeX/PDF mismatch) is editorial and the repository copy is already self-consistent.

The remaining items require independent editorial passes on [1]:

- **Item 4 (non-circular mosaic patchwork)**: Sections 5.4–5.6 of [1] should be reordered so that the abstract pair-label function and triple set are defined before the patchwork lemma is invoked, and a finite composition-table check on every abstract triple is required as a finite validity clause. No change to the v2 (M6') statements is needed.
- **Item 5 (overlap amalgamation)**: the overlap-amalgamation lemma should require a single cross-label assignment compatible with every overlap node and every mixed triple, not merely per-triple existence. This is a tightening of the lemma's hypothesis, orthogonal to (M6').
- **Item 6 (joint equality-witness compatibility as an explicit clause)**: the assumption (A4) of the typed-EQ quotient theorem (Section 7 of [1]) should be promoted to an explicit finite validity clause on the certificate. (V9.Eb') of §5 above is the natural per-stratum form; cross-stratum joint-witnessing is vacuous by (M6'.a) (no cross-stratum equality classes).
- **Item 7 (half-open cycle convention)**: the request-cycle lemma in Section 6.5 of [1] should be rewritten with a half-open $[i, j)$ convention and an explicit endpoint mapping. This is orthogonal to (M6').
- **Item 8 (global finite regularization)**: the no-automata regularization argument in Section 8 of [1] should be promoted to an explicit global lemma covering all infinite vertical rays uniformly. This is orthogonal to (M6').

Items 4–8 do not affect the (M6') repair and can be addressed in independent editorial passes. The v2 architecture of this paper is the natural setting for those passes, since each of them is local to a specific section of [1].

10 Summary

The repair is a recursive lift of the round-4 verticalization discipline. The only structural change is that the flat $\text{Vrt} \subseteq P$ of a mosaic is replaced by a finite *stratum forest* $\mathcal{V}$, in which selected DR/PO side-witnesses may themselves open nested local vertical strata. The repair:

- (1) closes the round-5 stress concept $\exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$ by accepting the small certificate of §7;
- (2) preserves Lemma 7.1 of [1] (vertical realisation of PP/PPI pairs) by the recursive case analysis of §4;
- (3) preserves the round-4 case analysis of Subcase (ii.e) of the Vertical-Realisation Lemma by restricting the analysis to one stratum at a time;
- (4) recovers the round-4 (M6) discipline as the depth-1 case of (M6');
- (5) preserves the rest of the round-4 architecture verbatim, modulo the per-stratum quantifier on (V3), (V4), (V5), (V9.Eb);
- (6) replaces the polynomial side-width $m(n) = O(n^3)$ with an effectively computable recurrence over modal depth, which suffices for decidability; the double-exponential certificate-space bound $B(C_0) \leq 2^{2^{p(n)}}$ is preserved.

The repair is computationally corroborated: all three Python reasoners in the repository (cover-tree tableau, baseline and cycle-aware quasimodel) report C_{side} as SAT. The v2 discipline (M6') accepts that satisfiable concept by construction.

References

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- [4] Michael Wessel. *Some Practical Issues in Building Description Logic Based Spatially-Aware Information Systems*. In: *Description Logics*, 2003.

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